

CUSTOMER

PROJECT

MANUFACTURER

DOCUMENT NO.

DOC TITLE

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Ampacity Calculation

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MV Cable Ampacity Calculation

Unit

Reference

Clause

Symbol

Formula

TREFOIL GND

TREFOIL DUCT

TREFOIL AIR

DESIGN DATA

Bonding Type		Design data			solid	solid	solid
Cond Mat		Design data			Al	Al	Al
Cond size	mm²	Design data			185	185	185
Cond design		Design data		CC	CC	CC	CC
Cond Dn	mm	Design data		Dc	15.9	15.9	15.9
Cond Sc Dn	mm	Design data		Ds	17.6	17.6	17.6
Ins Mat		Design data			XLPE	XLPE	XLPE
Ins Dn	mm	Design data		Di	33.9	33.9	33.9
Ins Sc Dn	mm	Design data		dx	37.6	37.6	37.6
Ins Tn	mm	Design data		Ti	9.19	9.19	9.19
Core Dn	mm	Design data		Do	--	--	--
Lay-up OD	mm	Design data		Da	--	--	--
Cable Dn	mm	Design data		De	43.4	43.4	43.4
Core No	nos	Design data		n	1	1	1
Wire No	nos	Design data		nw	37	37	37
Cable weight	kg/km	Design data		M	6731	6731	6731
Max cond temp	°C	Design Data		θmax	90	90	90
Earth fault current	A	Design Data		I _{sc} /1sec	1493	1493	1493
Rated voltage	kV	IEC60502-2	Table 1	U	33	33	33
Earth voltage	kV	IEC60502-2	Table 1	U ₀	19	19	19
Impulse voltage	kV	IEC60502-2	Table 14	Up	170	170	170

PARAMETER SETTING

Soil TR	K.m/W	Input		ρ	1.2	1.2	1.2	1.2
Multicore factor		Input		f _m	1	1	1	1
Frequency	Hz	Input		f	50	50	50	50
Ambient temp.	°C	Input		θ _a	30	30	40	40
Depth	mm	Input		L	1200	1000	1000	1000
Duct Type		Input			HDPE	HDPE	HDPE	HDPE
External diameter of duct	mm	Input		d _c	160	110.0	110.0	110.0
Thickness of duct	mm	Input		T _d	0	0	5.0	5.0
Duct temperature	°C	Input		θ _m	85	85	85	85
Core space factor		Input		f _s	1	1	1	1
Axial Circ distance H	mm	Input		S _h	500	0	0	0
Axial Circ distance V	mm	Input		S _v	500	0	0	0
Horizontal circuit	nos	Input		n _h	1	1	1	1
Vertical circuit	nos	Input		n _v	1	1	1	1
Temp. rise, ambient to max.	°C	IEC60287-1-1	1.4.1.1	Δθ	θ _{max} - θ _a	60	60	50
Core temperature, θ _c	°C	Iteration		θ _c		80	78	75
Axial Core distance	mm			S		44	160	44
Cable space	mm			S'		0	116	0
Circuit spacing H	mm			S' _h		0	0	0
Circuit spacing V	mm			S' _v		0	0	0
Axial distance cond to cab	mm			c		0	0	0
Ins Tn between cond	mm			t		0	0	0
Circumscribing radius cond	mm			r ₁		0.00	0.00	0.00
Coefficient belted cable		IEC60287-2	2.1.1.2	F ₂	1+3t/[2π(dx+t)-t]	0.00	0.00	0.00
Geometric factor		IEC60287-2	2.1.1.2	G	3F ₂ ln(da/2r ₁)	0.000	0.000	0.000
Wire factor		BICC	Table 2.1	f _w		0.0528	0.0528	0.0528
Temp coeff - conductor		IEC60287-1-1	Table 1	a ₂₀		0.004	0.004	0.004
Ins Relative permittivity		IEC60287-1-1	Table 3	ε _r		2.5	2.5	2.5
Loss factor		IEC60287-1-1	Table 3	tanδ		0.004	0.004	0.004
Dielectric loss	W/m/ph	EC60287-1-1	2.2	W _d	ωCU ⁰ tanδ	0.000	0.000	0.000

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MV Cable Ampacity Calculation								
	Unit	Reference	Clause	Symbol	Formula	TREFOIL GND	TREFOIL DUCT	TREFOIL AIR

DC RESISTANCE

DC resistant at 20degC	ohm/m	IEC60228 Class 2		R0		1.640E-04	1.640E-04	1.640E-04
DC resistant at ambient	ohm/m	IEC60287-1-1	2.1.1	R0amb	$R0(1+a20(Ox-20))$	1.706E-04	1.706E-04	1.706E-04
DC resistant at 90degC	ohm/m	IEC60287-1-1	2.1.1	R0max	$R0(1+a20(Omax-20))$	2.103E-04	2.103E-04	2.103E-04

AC RESISTANCE

Strand constant		IEC60287-1-1	Table 2	ks		1.0	1.0	1.0
Strand constant		IEC60287-1-1	Table 2	kp		0.8	0.8	0.8
Spacing correction		Mc Graw Hill	Page 161			1.250	1.250	1.250
Bessel function		IEC60287-1-1	2.1.2	Xs4	$[ks(8pf/R0)10-7]2$			
Skin effect		IEC60287-1-1	2.1.2	Ys	$Xs4/(192+0.8Xs4)$			
Proximity function		IEC60287-1-1	2.1.4	Xp4	$[kp(8pf/R0)10-7]2$			
Proximity function		IEC60287-1-1	2.1.4	Yp"	$Xp4/(192+0.8Xp4)$			
Proximity effect		IEC60287-1-1	2.1.4	Yp	$Yp"(Dc/s)2[0.312(Dc/s)2+1.18/(Yp"+0.27)]$			
AC resistant at 20degC	ohm/m	IEC60287-1-1	2.1	R20	$R0(1+Ys+Yp)$			
AC resistant at 20degC - isolated	ohm/m	IEC60287-1-1	2.1	R'20	$R0(1+Ys)$			
AC resistant at 90degC	ohm/m	IEC60287-1-1	2.1	R90	$R0max(1+Ys+Yp)$	2.108E-04	2.107E-04	2.108E-04
AC resistant at 90degC - isolated	ohm/m	IEC60287-1-1	2.1	R'90	$R0max(1+Ys)$			

SHEATH LOSS FACTOR

Equivalent mean diameter	mm	Design data		ds				
Equivalent resistant at 20degC	ohm/m	IEC60287-1-1		R20	p/A	3.448E-04	3.448E-04	3.448E-04
Equivalent resistant at ambient	ohm/m	IEC60287-1-1		Rx	$R20(1+a20(Ox-20))$			
Equivalent resistant at Oc	ohm/m	IEC60287-1-1		Rs	$R20(1+a20(Oc-20))$			
Sheath Inductance	H/m	IEC60287-1-1	2.3.3	L	$2ln(2s/Dm)x10-7$			
Sheath Reactance	ohm/m	IEC60287-1-1	2.3.3	X	$2wln(2s/Dm)x10-7$			
Circulating current loss factor		IEC60287-1-1	2.3.1	lam1'	$Rs/R/(1+(Rs/X)2)$			
Eddy current loss factor		IEC60287-1-1	2.3.6.1	lam1"		0.000E+00	0.000E+00	0.000E+00
Sheath loss factor		IEC60287-1-1	2.3	lam1	$lam1'+lam1"$	0.033	0.134	0.033

MV Cable Ampacity Calculation

Unit	Reference	Clause	Symbol	Formula	TREFOIL GND	TREFOIL DUCT	TREFOIL AIR
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THERMAL RESISTANCE

Thermal Resistant Insulation	K.m/W	IEC60287-2-1	T1	$(p/2p)LN[1+(2t/dc)]$	0.451	0.451	0.451
Thermal Resistant filler	K.m/W	IEC60287-2-1	T2f		0.000E+00	0.000E+00	0.000E+00
Thermal Resistant Bedding	K.m/W	IEC60287-2-1	T2		0.000E+00	0.000E+00	0.000E+00
Thermal Resistant Outercovering	K.m/W	IEC60287-2-1	T3	$(p/2p)LN[1+(2t/D)]$	0.208	0.130	0.130
Thermal Resistant External	K.m/W	IEC60287-2-1 2.2	T4		2.231	2.377	1.011

CURRENT RATING

Critical Temp Rise of Soil	degC	IEC60287-1-1	dO	$Wk(pc/2p)LN(Gf)$	0.000E+00	0.000E+00	0.000E+00
Top Equation		IEC60287-1-1 1.4.1	Etop	$dO-Wd(0.5T1+n(T2+T3)+T4)-dO$	59.9	59.9	50.0
Bottom Equation		IEC60287-1-1 1.4.1	Ebot	$RT1+nR(1+l\text{am}1)T2+nR(1+l\text{am}1+l\text{am}2)(T3+T4)$	36.24E-04	6.939E-04	3.438E-04
Calculated Amps	A	IEC60287-1-1 1.4.1	I	$\text{sqrt}(Etop/Ebot)$	309	294	381
Calculated MVA	MVA	IEC60287-1-1 1.4.1	MVA	$\text{sqrt}3*U*I$	18	17	22

ELECTRICAL DATA

Inductance	uH/m	IEC60287-1-1 2.3	L	$0.2LN(2s/Dm)$			
Reactance	uohm/m	IEC60287-1-1 2.3	X	$w*Lm*10^{-3}$			
Pos Seq Resistance	uohm/m		R1	$R'1(1+l\text{am}1+l\text{am}2)$			
Pos Seq Reactance	uohm/m		X1	$w*L*10^{-3}$			
Pos Seq Impedance	uohm/m		Z1	$\text{sqrt}(R12+Xm2)$			
Zero Seq Resistance	uohm/m		R0	$Rx+(n*R'x)$			
Capacitance	uF/km	IEC60287-1-1 2.2	C	$e/(18LN(Di/ds))$	2.119E-10	2.119E-10	2.119E-10
Charging current - Max	A/km	BICC Page 170	Ic max	$1.08*w*C*U0$			

THERMAL LOSS

Thermal Resistant Total	K.m/W	Mc Graw hill	Page 34	SumT	$T1+T2+T3+T4$		
Thermal loss	W/m	Mc Graw hill	Page 34	W	$(Oc-Oa)/\text{SumT}$		
Losses - Conductor	W/m	IEC60287-1-1 2.4.2	Wc	$3(I2R)$	20.18	18.20	30.64
Losses - Dielectric	W/m	IEC60287-1-1 2.2	Wd	$3*w*C*U02*\text{tand}$	0.000E+00	0.000E+00	0.000E+00
Losses - Metal sheath	W/m	IEC60287-1-1	Wm	$Wc*l\text{am}1$			

INDUCED PARAMETER

Normal - Sheath voltage	V/m	IEEE Std 575	Page 29	Us	$I*X$		
Normal - Circulating current	A	IEEE Std 575	Page 29	Is	$\text{sqrt}(I2(R/Rs)l\text{am}1')$		
Transient - Sheath voltage	V/m	IEEE Std 575	Page 29	Ust	$Ix*X$		
Transient - Circulating current	A	IEEE Std 575	Page 29	Ist	$\text{sqrt}(Ix2(R/Rs)l\text{am}1')$	None	None

VOLTAGE DROP

Rated voltage	kV	IEC60840	Table 4	U		33	33	33
Pos Seq Impedance	uohm/m			Z1	$\text{sqrt}(R12+Xm2)$			
Calculated Amps	A	IEC60287-1-1 1.4.1	I	$\text{sqrt}(Etop/Ebot)$	309	294	381	
Permissible limit	%	Input	a		2.5%	2.5%	2.5%	
Permissible voltage drop	V		E	$a*U$	825	825	825	
Voltage drop factor	mV/A/m		R	$\text{sqrt}3*Z$				
Max permissible length	km		L	$E/(I*R)$				

CRITICAL LENGTH

Critical Length	km		Lmax	$\text{sqrt}3(I/wCU)x103$	None	None	None	
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RECEIVED POWER

Max Feeding Power	MVA	IEC60287-1-1 1.4.1	S					
Max Received Power	MW		P	$\text{sqrt}[S2-Q2]$				
Power Losses	%		Ploss	P/S	%	%	%	
